

CS101

Boolean Logic

2/10

Spot 3 Differences

```
public class Calculator {  
    public static void main(String[] args) {  
        int balance = 1000;  
        int deposit = 250;  
        int withdrawal = 75;  
  
        // Update balance  
        balance += deposit;  
        balance -= withdrawal;  
  
        // Calculate fee (2% of balance)  
        double fee = balance * 0.02;  
  
        // Check if divisible by 100  
        int remainder = balance % 100;  
  
        System.out.println("Balance: $" + balance);  
        System.out.println("Fee: $" + fee);  
    }  
}
```

```
public class Calculator {  
    public static void main(String[] args) {  
        int balance = 1000;  
        int deposit = 250;  
        int withdrawal = 75;  
  
        // Update balance  
        balance += deposit;  
        balance -= withdrawal;  
  
        // Calculate fee (2% of balance)  
        double fee = balance x 0.02;  
  
        // Check if divisible by 100  
        int remainder = balance % 100;  
  
        System.out.println("Balance: $" + balance);  
        System.out.println("Fee: $" + fee);  
    }  
}
```

Agenda

1. Demo, programmerVernacular
2. Alex Lee + w3 Tutorials (as group)
3. Work time
4. Quiz4 Practice (Kahoot)
5. Touch Typing

Announcements

1. No Class 2/17
2. No Quiz 2/19
3. Extended labs to Friday nights
4. CS101 GitHub [README.md](#) reversed Topic order

Demo

programmerVernacular

Alex Lee Boolean Tutorial

<https://www.youtube.com/watch?v=CHVVEGRGiJU>

W3 if/else Tutorial

https://www.w3schools.com/java/java_conditions.asp

Java Comparison Operators

<i>op</i>	<i>meaning</i>	<i>true</i>	<i>false</i>
==	<i>equal</i>	2 == 2	2 == 3
!=	<i>not equal</i>	3 != 2	2 != 2
<	<i>less than</i>	2 < 13	2 < 2
<=	<i>less than or equal</i>	2 <= 2	3 <= 2
>	<i>greater than</i>	13 > 2	2 > 13
>=	<i>greater than or equal</i>	3 >= 2	2 >= 3

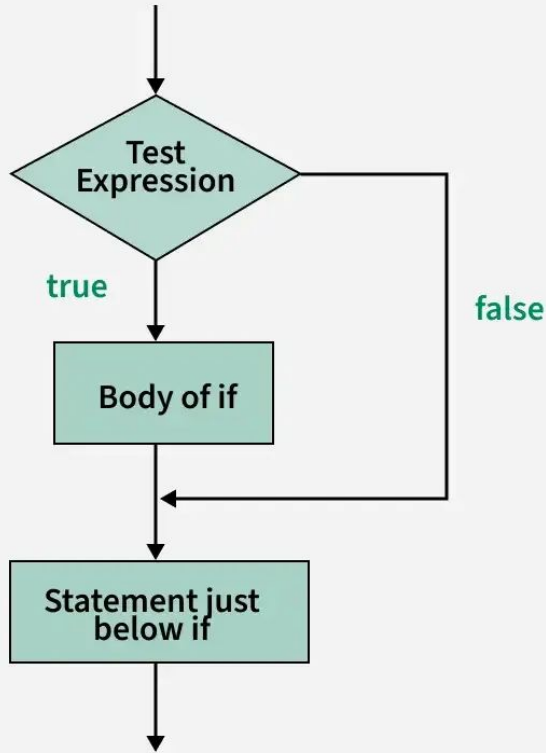
Java Logical Operators

<i>op</i>	<i>meaning</i>	<i>true</i>	<i>false</i>
&&	<i>AND (both true)</i>	<code>true && true</code>	<code>true && false</code>
 	<i>OR (at least one true)</i>	<code>true false</code>	<code>false false</code>
!	<i>NOT (inverts boolean)</i>	<code>!false</code>	<code>!true</code>

Truth Table for Logical Operators

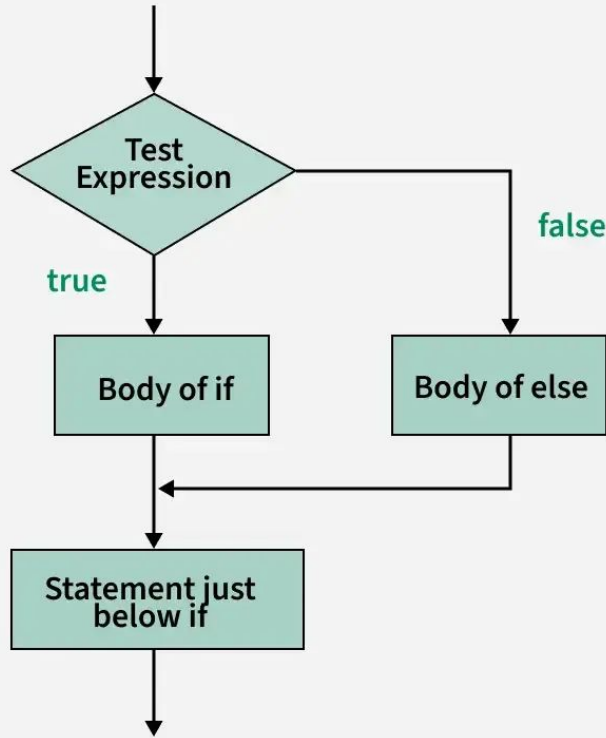
<i>A</i>	<i>B</i>	<i>A && B</i>	<i>A B</i>
true	true	true	true
true	false	false	true
false	true	false	true
false	false	false	false

Control Flow Diagram: “if” statement



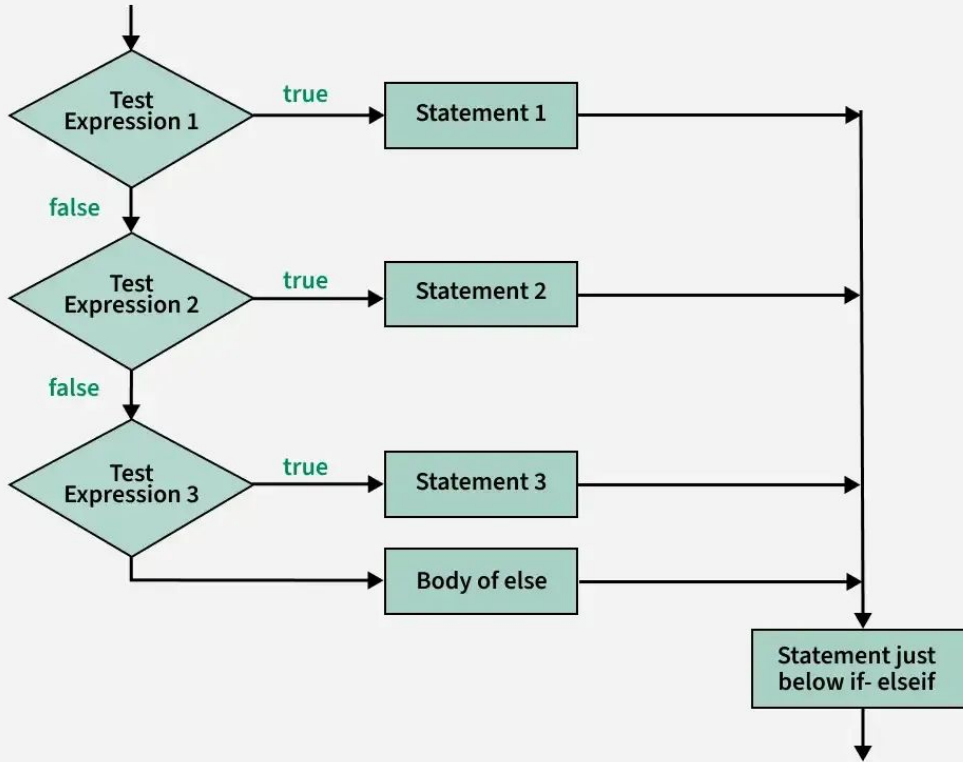
```
int x = 4;  
if (x > 10) {  
    // Run this code if `true`  
}
```

Control Flow Diagram: “if/else” statement



```
int x = 4;  
if (x > 10) {  
    // Run this code if `true`  
} else {  
    // Run this code if `false`  
}
```

Control Flow Diagram: “if/elseif” ladder



```
int x = 4;

if (x > 10) {
    // Run this code if `true`
    System.out.println("x > 10");
} else if (x > 5){
    // Run if first condition false, but this is true
    System.out.println("x > 5 and x <= 10");
} else if (x > 0){
    // Run if both above false, but this is true
    System.out.println("x > 0 and x <= 5");
} else {
    // Run if all above false
    System.out.println("x <= 0");
}
```

Before Next Class:

- Start HW2! Short demo and time for Qs next class
- Submit Lab1 and Lab2, due FRIDAY!!
- Due dates: canvas.arcadia.edu:

Next Class 2/12

1. Lab5, Booleans and Conditionals

- **Bring Headphones every class!**